

## SAFETY DATA SHEET

**Creation Date** 26-September-2009

**Revision Date** 05-September-2023

**Revision Number** 2

### 1. Identification

**Product Name** 4-Isocyanato-4-thien-2-yltetrahydropyran

**Cat No. :** CC62206CB; CC62206DA; CC62206ZZ

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Fisher Scientific  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                           |            |
|---------------------------|------------|
| Acute oral toxicity       | Category 4 |
| Acute dermal toxicity     | Category 4 |
| Acute Inhalation Toxicity | Category 4 |
| Respiratory Sensitization | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled

May cause allergy or asthma symptoms or breathing difficulties if inhaled

Harmful if inhaled

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
If experiencing respiratory symptoms: Call a POISON CENTER/doctor  
Take off contaminated clothing and wash it before reuse

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                | CAS-No      | Weight % |
|------------------------------------------|-------------|----------|
| 4-Isocyanato-4-thien-2-yltetrahydropyran | 926921-63-3 | > 95     |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                                    |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                                                          |
| <b>Inhalation</b>                      | Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention.                                               |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                     |
| <b>Most important symptoms/effects</b> | May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                              |

### 5. Fire-fighting measures

|                                     |                                                                              |
|-------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
|-------------------------------------|------------------------------------------------------------------------------|

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                  |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                     |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Do not let this chemical enter the environment. Avoid dust formation. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Avoid contact with skin and eyes. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Minimize dust generation and accumulation. Wash hands before breaks and immediately after handling the product. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep refrigerated. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases. Strong reducing agents.                                                                                                                               |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.                                                                                                                                                                                                                       |
| <b>Engineering Measures</b>          | Ensure adequate ventilation, especially in confined areas.<br>Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |
| <b>Personal protective equipment</b> |                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Eye Protection</b>                | Goggles                                                                                                                                                                                                                                                                                                                                                            |

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                                   |
|-----------------------------------------------|-----------------------------------|
| <b>Physical State</b>                         | Solid                             |
| <b>Appearance</b>                             | Light yellow                      |
| <b>Odor</b>                                   | No information available          |
| <b>Odor Threshold</b>                         | No information available          |
| <b>pH</b>                                     | No information available          |
| <b>Melting Point/Range</b>                    | 49.5 - 52.5 °C / 121.1 - 126.5 °F |
| <b>Boiling Point/Range</b>                    | No information available          |
| <b>Flash Point</b>                            | No information available          |
| <b>Evaporation Rate</b>                       | No information available          |
| <b>Flammability (solid,gas)</b>               | No information available          |
| <b>Flammability or explosive limits</b>       |                                   |
| Upper                                         | No data available                 |
| Lower                                         | No data available                 |
| <b>Vapor Pressure</b>                         | No information available          |
| <b>Vapor Density</b>                          | No information available          |
| <b>Specific Gravity</b>                       | No information available          |
| <b>Solubility</b>                             | No information available          |
| <b>Partition coefficient; n-octanol/water</b> | No data available                 |
| <b>Autoignition Temperature</b>               | No information available          |
| <b>Decomposition Temperature</b>              | No information available          |
| <b>Viscosity</b>                              | No information available          |
| <b>Molecular Formula</b>                      | C10 H11 N O2 S                    |
| <b>Molecular Weight</b>                       | 209.27                            |

## 10. Stability and reactivity

|                            |                                              |
|----------------------------|----------------------------------------------|
| <b>Reactive Hazard</b>     | None known, based on information available   |
| <b>Stability</b>           | Stable under normal conditions.              |
| <b>Conditions to Avoid</b> | Incompatible products. Avoid dust formation. |

|                                         |                                                                                                            |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Strong bases, Strong reducing agents                                |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NO <sub>x</sub> ), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                              |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

### **Component Information**

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** May cause skin, eye, and respiratory tract irritation

**Sensitization** May cause sensitization by inhalation

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------------------------|-------------|------------|------------|------------|------------|------------|
| 4-Isocyanato-4-thien-2-yltetrahydropyran | 926921-63-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

## 13. Disposal considerations

**Waste Disposal Methods**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT**

UN-No UN2811  
 Proper Shipping Name Toxic solid, organic, n.o.s.  
 Technical Name 4-Isocyanato-4-thien-2-yltetrahydropyran  
 Hazard Class 6.1  
 Packing Group III

**TDG**

UN-No UN2811  
 Proper Shipping Name Toxic solid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

**IATA**

UN-No UN2811  
 Proper Shipping Name TOXIC SOLID, ORGANIC, N.O.S.\*  
 Hazard Class 6.1  
 Packing Group III

**IMDG/IMO**

UN-No UN2811  
 Proper Shipping Name Toxic solid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

## 15. Regulatory information

**International Inventories**

| Component                                | CAS-No      | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|------------------------------------------|-------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| 4-Isocyanato-4-thien-2-yltetrahydropyran | 926921-63-3 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                                | CAS-No      | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------|-------------|-------|------|------|------|------|------|-------|-------|
| 4-Isocyanato-4-thien-2-yltetrahydropyran | 926921-63-3 | -     | -    | -    | -    | -    | -    | -     | -     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

**Authorisation/Restrictions according to EU REACH**

Not applicable

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                | CAS-No      | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------------------|-------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 4-Isocyanato-4-thien-2-yltetrahydropyran | 926921-63-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                | CAS-No      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------------------|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 4-Isocyanato-4-thien-2-yltetrahydropyran | 926921-63-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

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**Creation Date** 26-September-2009

**Revision Date** 05-September-2023

**Print Date** 05-September-2023

**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**